

Demonstrating Eyes-Free Object Retrieval via Fingertip Deflection Guidance Using the NURing

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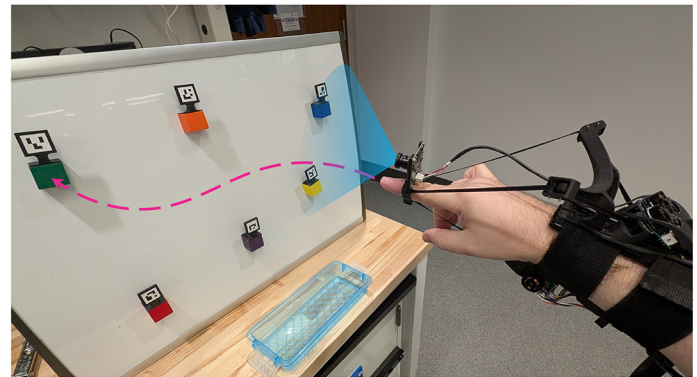
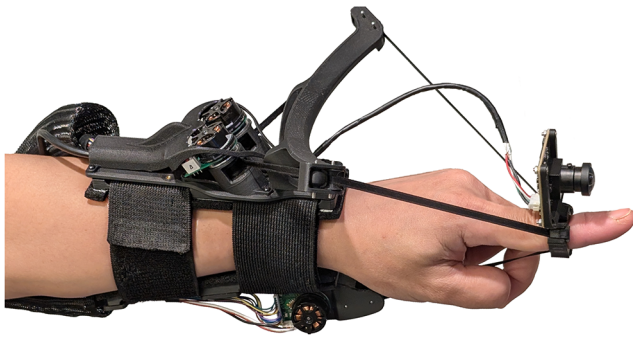


Figure 1: The NURing Device (A) as worn by the user and (B) providing guidance cues during an eyes-free retrieval task.

Abstract

Many eyes-free guidance systems convey directional information through symbolic cues requiring interpretation, potentially increasing cognitive load and competing with existing sensory channels. We present an interactive demonstration of fingertip deflection as a continuous, physically-intuitive guidance cue that gently biases the arm during reach. Motivated by a challenge frequently reported by individuals with blindness — that of safely retrieving small objects within arm's reach — we developed an eyes-free, object-retrieval task. Building on our prior NURing, we extend fingertip deflection beyond instrumented testbeds by integrating a camera on the actuation ring. This allows the NURing to detect ArUco-tagged objects, estimate their pose relative to the camera, and drive continuous closed-loop deflection cues toward the target in real time. This demonstration invites attendees to experience this embodied guidance firsthand and to explore how fingertip deflection could

support future assistive and collaborative systems that guide action through physical intuition rather than cognitive translation.

CCS Concepts

• **Human-centered computing** → **Haptic devices**; *Accessibility systems and tools*.

Keywords

Haptics, Wearable, Accessibility, Guidance, Individuals with Blindness or Visual Impairment, Assistive Technologies

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1 Introduction

For individuals with blindness or visual impairment (BVI), interactions requiring the precise location and manipulation of small objects in the peripersonal space are often difficult or infeasible.

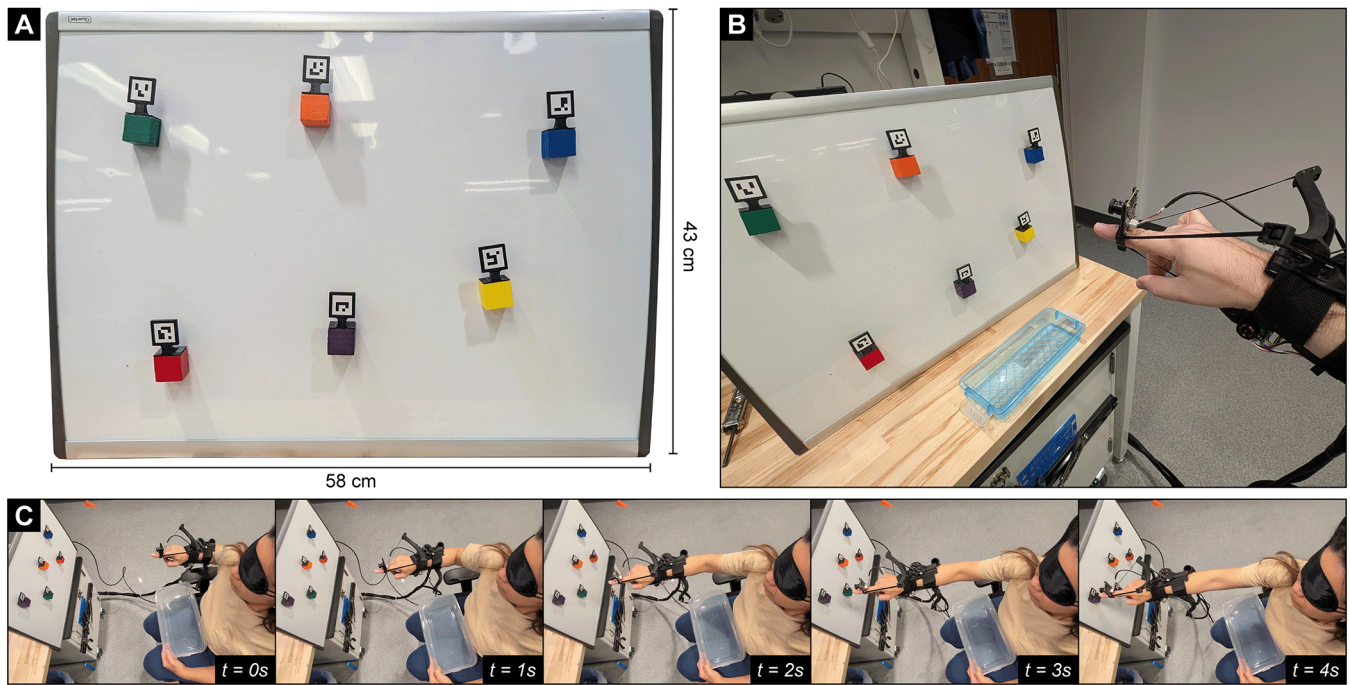


Figure 2: (A) Components of the interactive demo experience (magnetic object board with ArUco-tagged objects), (B) Image of NURING generating deflection towards the top-left green target object, and (C) extracted video frames showing continuous guidance to target over four-second trajectory.

Safely retrieving small items, such as dropped pills or kitchen utensils, is frequently reported as one of the most time-consuming and frustrating challenges for individuals with BVI [9]. Yet, these situations are commonplace and often safety-critical, particularly in living environments shared with children or pets. Over time, this inability to easily perform everyday tasks can undermine independence and agency, and has been shown to negatively impact mental health, contributing to anxiety or depression [1].

In an attempt to enable non-visual interaction, prior work has often shifted guidance information to other sensory channels. Audio-based approaches use directional tones, spoken commands, or spatialized sound to steer the user towards a target [10, 13, 20]. While effective in controlled settings, these cues can obscure an already critical sensory channel [15] and demand sustained attention, particularly in noisy environments or when targets move [6]. Many haptic approaches have relied on vibrotactile feedback delivered to the wrist, hand, or fingertips [2, 4, 12, 14] through modulated vibrations or coded patterns. Although these cues leave hearing unobstructed, they are often perceived as alerts rather than continuous guidance cues [5], and may require extensive training or cognitive interpretation before action, leading to slower or more hesitant movements [3, 7]. Moreover, variations in tactile sensitivity can increase ambiguity [17], and wearable vibrotactile elements can interfere with natural touch and object manipulation [16].

Despite decades of research into assistive guidance technologies, it is telling that the most prevalent guidance aid for individuals with BVI remains the white cane [8]. Its continued use reflects a key principle: the cane provides continuous, direct, and naturally

understood physical feedback about the environment, allowing users to feel what is happening rather than interpret symbolic cues. This suggests that non-visual guidance is most effective when it supports action through direct, embodied cues with continuous feedback.

Motivated by this principle, we explored fingertip deflection as a guidance cue, inspired by the familiar act of a child leading an adult by the finger [11]. Rather than encoding commands into signals that must be decoded, fingertip deflection gently biases the user's hand motion during reach, providing a directional cue that is continuous, immediately interpretable, and preserves user agency.

Our prior work validated fingertip deflection as generated by the NURING device for guiding users towards target poses along single-axis [18] and dual-axis [19] directions using a controlled, sensorized testbed that constrained motion to pivot about the elbow.

The work presented here advances fingertip deflection guidance to real-world object interaction by integrating sensing directly into the wearable through a camera mounted on the ring. This camera estimates target location from ArUco-tagged objects and enables real-time guidance during reach. This sensing modality also provides a clear path towards future integration of markerless scene and object recognition guidance. The interactive demonstration presented here allows attendees to directly experience the continuous, embodied guidance produced by coupling fingertip deflection with ring-mounted vision, guiding reach towards physical target objects without visual attention, audio prompts, or symbolic decoding.

2 Interactive Demo: Eyes-Free Object Retrieval

Our interactive demonstration centers on an eyes-free object retrieval task inspired by challenges described by members of the BVI community. One recurring theme was that of safely retrieving small objects that had been dropped to the floor, such as pills or utensils. If not retrieved, such objects can pose hazards for children or pets, motivating the need for effective guidance mechanisms that do not rely on vision.

In our task, attendees interact with a vertical object board positioned at arm's length (approximately 70 centimeters). Five small objects (25-mm cubes), each with a compact (24-mm) ArUco tag, are distributed randomly across the board (Fig. 2A). The NURing's onboard camera detects these tags and estimates their position in real time (Fig. 2B), allowing the controller to generate continuous fingertip deflection cues that guide the attendee's arm towards the selected object (Fig. 2C).

To prepare, attendees are seated to ensure comfort and accessibility and are provided with a small tray placed on their lap for depositing retrieved objects. They select an actuation ring that fits comfortably on the index finger, just above the proximal interphalangeal joint. The facilitator secures the forearm brace and drive band and applies a light baseline tendon tension to stabilize the ring. Then, the facilitator introduces the fingertip deflection cue, allowing attendees to feel how deflection of the fingertip influences arm motion.

Before starting eyes-free guidance, attendees are invited to retrieve one or two objects with their eyes open. This brief familiarization will allow them to establish a sense of reach distance and object shape. Once ready, attendees close their eyes and raise their arm towards the interaction board to begin the guided task.

At the start of each trial, the facilitator will select one object as the active target. As the attendee moves their arm forward, the NURing provides continuous fingertip deflection cues that steer the hand towards the target object. Upon contact, the attendee grasps the object, places it in the tray, and returns their hand to the initial position. The facilitator selects the next target, and the guidance process repeats until all five objects have been retrieved.

After completing the task, attendees are given the opportunity to share feedback and suggestions on a nearby laptop while the system is reset for the next demonstration. Attendees should be able to complete the demonstration in under five minutes, with the retrieval task itself taking under a minute.

3 The NURing Device

As the core device design has been detailed in our prior work [18, 19], here we only summarize the high-level device features before introducing the new ring-mounted camera integration that drives closed-loop fingertip deflection guidance during reach.

3.1 Device Overview

The NURing is a tendon-driven wearable device (Fig. 3A) that generates directional fingertip deflection using three independently-actuated tendons routed from a forearm-mounted drive band to a lightweight actuation ring worn on the index finger, just above the proximal-interphalangeal (PIP) joint. Coordinated tendon actuation enables continuous deflection cues across the anatomical directions

(abduction, adduction, flexion, and extension, Fig. 3B), allowing the system to bias the user's hand toward a given direction during the reaching motion. By imparting these deflections near the PIP joint, the fingertip pad remains unobstructed, allowing for natural touch sensing and dexterous manipulation. Care was given to ensure user comfort, with actuators mounted on a padded forearm brace, and thin polyester ribbons for tendons to minimize friction where skin-tendon interaction may occur. During use, the device maintains a small holding tension to keep the ring seated and stable. Deflection cues are perceptible but intentionally limited both in force and magnitude, preserving agency by allowing users to override cues as well as preventing deflection outside of the user's active range of motion (for safety/comfort). The wearable device weighs approximately 270 grams, similar to a smartphone.

3.2 Computer Vision Integration

In our previous NURing studies [18, 19], closed-loop guidance relied on external sensing of forearm orientation measured within an instrumented platform. New in this work is the integration of a wide-angle global-shutter camera (120-degree field of view), mounted on the actuation ring to provide a fingertip-centric view aligned with the guided hand. This new sensing modality enables guidance during free reach towards objects rather than motion constrained to a pivoting testbed. The camera captures 1600x1200-pixel frames at 60 Hz, enabling responsive closed-loop guidance during reach. Frames are processed and undistorted using an intrinsic camera calibration model to correct for lens distortion and improve pose estimation stability and accuracy.

We use OpenCV to detect fiducial markers present in the frame using the ArUco library. For each detected ArUco tag, the system estimates the tag's 3D translation in physical units relative to the camera, resulting in a target position in the camera's coordinate frame. Because the camera is mounted to the ring and therefore co-located with the finger, the measured displacement between the target and the camera's principal point provides a direct measure of the direction the hand should move towards to reach the target (Fig. 3C).

This measured target displacement provides a direct error vector for closed-loop guidance. At each frame, the system computes the current error vector towards the target in the camera's coordinate frame, and then maps the vector heading to a deflection direction. The proportional controller then scales the magnitude of this deflection and computes the individual motor contributions needed to impart coordinated fingertip deflection in the direction of the target. These deflections bias the user's ongoing arm motion, allowing the user to converge on the object through continuous, direct embodied guidance rather than symbolic interpretation.

Should the user lose tracking of the intended target, the system can generate a salient "stop" cue by pulling on all three tendons at once, rendering the illusion of hitting a virtual wall and indicating that the user should reverse their motion. Alternatively, through early studies we have found that rapidly pulling then releasing all three tendons simultaneously over a short interval generates a very strong alert cue, causing users to stop forward motion and pull back their hand to re-acquire the target. Ongoing work will explore

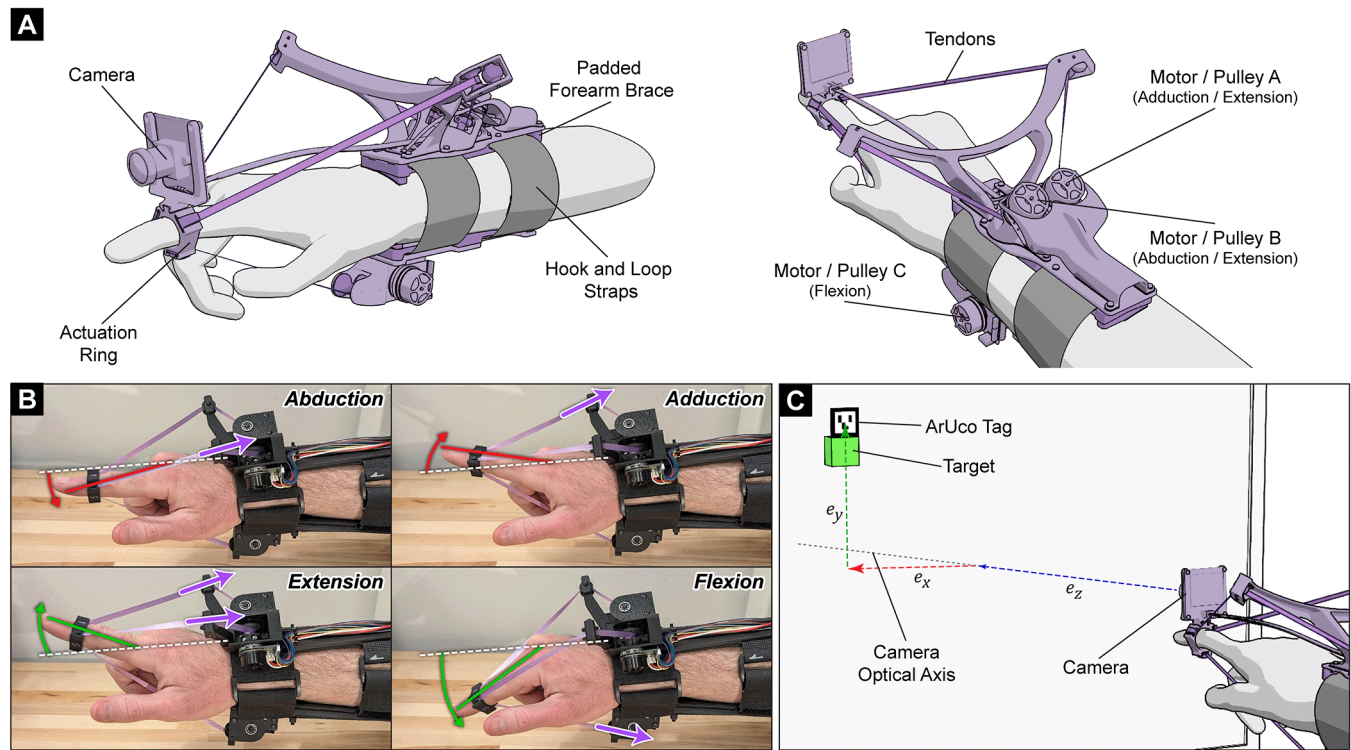


Figure 3: (A) Components of the NURing device, (B) fingertip deflection in the 4 anatomical directions (adapted from [19]), and (C) ArUco tag translational information serves as error inputs for the guidance controller.

further haptic effects that can be imparted by fingertip deflection to signify other cues.

While ArUco tags provide a robust and repeatable target representation for the interactive demonstration, the ring-mounted camera establishes a starting point for future markerless sensing. In ongoing work, the same pipeline could be extended to use standard computer vision methods (e.g., object recognition, scene understanding, SLAM) to localize targets and guide interactions without predefined fiducial markers.

4 Conclusion

In this paper, we present fingertip deflection cues rendered by the NURing for intuitive, whole-arm guidance. We extend our previous work beyond instrumented testbeds by integrating a camera mounted on the actuation ring, enabling real-time target localization from ArUco-tagged objects for closed-loop guidance during reach. In our interactive demonstration described here, attendees will be able to directly experience these fingertip deflection cues for continuous, eyes-free guidance towards a physical target object. By shaping arm motion through direct, embodied cues without relying on vision or occupying other sensory channels, this work highlights fingertip deflection as a practical guidance modality for real-world interaction, supporting future assistive and collaborative systems that guide action through intuition rather than symbolic, cognitive translation.

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